

Experiment No: 5

Title:

Application of Array Methods and Object Handling by Creating a Cart Total Calculator with Discount Logic.

Software/Tools Required:

- Visual Studio Code
- Web Browser such as Google Chrome
- HTML, CSS, and JavaScript

Theory:

JavaScript provides **arrays** and **objects** as powerful data structures for storing and managing data. An **array** is used to store multiple values in a single variable, while an **object** stores data as key-value pairs, making it suitable for representing real-world entities such as products, customers, or shopping cart items.

JavaScript offers several built-in **array methods** that simplify data processing. Methods such as `push()` are used to add elements, `pop()` removes the last element, `filter()` selects elements based on a condition, `map()` transforms array elements, `find()` searches for a specific element, `forEach()` iterates through all elements, and `reduce()` performs calculations such as finding the total price of items in a shopping cart.

Objects allow related information to be grouped together. For example, a product object may contain properties such as **name**, **price**, **quantity**, and **category**. Combining arrays and objects makes it possible to manage multiple products efficiently and perform operations such as adding items, calculating totals, and applying discounts.

In this experiment, a **Cart Total Calculator** is developed using JavaScript. Each product is represented as an object and stored inside an array. The user enters product details such as name, price, and quantity. JavaScript processes the array using appropriate array methods to calculate the subtotal of all products. Based on predefined conditions, a discount is applied to the total amount, and the final payable amount is displayed dynamically on the webpage.

Result/Conclusion:

Thus, JavaScript arrays, objects, and array methods were successfully implemented to create a Cart Total Calculator. The application efficiently managed product data, calculated the total bill, applied the discount logic, and displayed the final payable amount dynamically.

